IT Certification & Technical Interview Practice Exams

56 Topics × 30 Questions = 1,680 Questions

Each topic contains 10 Beginner, 10 Intermediate, and 10 Advanced questions. Answer keys follow each tier with the correct answer and a brief explanation.

1. Deep Learning & PyTorch

Category: AI & ML **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In Deep Learning & PyTorch, which statement best describes a core concept?

A. The primary mechanism used by Deep Learning & PyTorch

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with Deep Learning & PyTorch?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a Deep Learning & PyTorch component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

4. Which design goal is most important for a production Deep Learning & PyTorch solution?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

5. What is a common reason to automate Deep Learning & PyTorch operations?

A. Consistency and repeatability

B. To eliminate all testing

C. To hide failures

D. To prevent versioning

6. In Deep Learning & PyTorch, which statement best describes a core concept is the best choice?

A. The primary mechanism used by Deep Learning & PyTorch

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

7. Which practice is generally recommended when working with Deep Learning & PyTorch?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

8. A production issue occurs in a Deep Learning & PyTorch component. What should you check first? Choose the most appropriate option.

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

9. Which design goal is most important for a production Deep Learning & PyTorch solution in a troubleshooting context?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

10. What is a common reason to automate Deep Learning & PyTorch operations?

A. Consistency and repeatability

B. To eliminate all testing

C. To hide failures

D. To prevent versioning

Beginner Level Answer Key & Explanations

1. A — The first option represents a central concept used in Deep Learning & PyTorch. The other options are distractors unrelated to the core mechanism.

2. A — Versioning, testing, least privilege, and observability are broadly applicable engineering practices.

3. A — Start with evidence and recent changes before making disruptive modifications.

4. A — Production systems should balance reliability, security, maintainability, and observability.

5. A — Automation reduces manual variation and makes repeatable operations easier to test and audit.

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Intermediate Level (10 Questions)

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B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

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Advanced Level (10 Questions)

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2. LLM & OpenAI APIs

Category: AI & ML **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In LLM & OpenAI APIs, which statement best describes a core concept?

A. The primary mechanism used by LLM & OpenAI APIs

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with LLM & OpenAI APIs?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a LLM & OpenAI APIs component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

4. Which design goal is most important for a production LLM & OpenAI APIs solution?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

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5. What is a common reason to automate LLM & OpenAI APIs operations?

A. Consistency and repeatability

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6. In LLM & OpenAI APIs, which statement best describes a core concept is the best choice?

A. The primary mechanism used by LLM & OpenAI APIs

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A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

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8. A production issue occurs in a LLM & OpenAI APIs component. What should you check first? Choose the most appropriate option.

A. Logs, metrics, recent changes, and reproducible symptoms

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C. Disable security controls

D. Randomly change configuration

9. Which design goal is most important for a production LLM & OpenAI APIs solution in a troubleshooting context?

A. Reliability, security, maintainability, and observability

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10. What is a common reason to automate LLM & OpenAI APIs operations?

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Intermediate Level (10 Questions)

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3. Machine Learning & AI

Category: AI & ML **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In Machine Learning & AI, which statement best describes a core concept?

A. The primary mechanism used by Machine Learning & AI

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C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with Machine Learning & AI?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

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3. A production issue occurs in a Machine Learning & AI component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

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4. Which design goal is most important for a production Machine Learning & AI solution?

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Intermediate Level (10 Questions)

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Advanced Level (10 Questions)

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Advanced Level Answer Key & Explanations

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4. GraphQL & API Design

Category: APIs & Integration **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In GraphQL & API Design, which statement best describes a core concept?

A. The primary mechanism used by GraphQL & API Design

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with GraphQL & API Design?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a GraphQL & API Design component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

4. Which design goal is most important for a production GraphQL & API Design solution?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

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5. What is a common reason to automate GraphQL & API Design operations?

A. Consistency and repeatability

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A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

9. Which design goal is most important for a production GraphQL & API Design solution in a troubleshooting context?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

10. What is a common reason to automate GraphQL & API Design operations?

A. Consistency and repeatability

B. To eliminate all testing

C. To hide failures

D. To prevent versioning

Beginner Level Answer Key & Explanations

1. A — The first option represents a central concept used in GraphQL & API Design. The other options are distractors unrelated to the core mechanism.

2. A — Versioning, testing, least privilege, and observability are broadly applicable engineering practices.

3. A — Start with evidence and recent changes before making disruptive modifications.

4. A — Production systems should balance reliability, security, maintainability, and observability.

5. A — Automation reduces manual variation and makes repeatable operations easier to test and audit.

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Intermediate Level (10 Questions)

1. In GraphQL & API Design, which statement best describes a core concept?

A. The primary mechanism used by GraphQL & API Design

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with GraphQL & API Design?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

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Advanced Level (10 Questions)

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5. Django & Flask

Category: Backend **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In Django & Flask, which statement best describes a core concept?

A. The primary mechanism used by Django & Flask

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with Django & Flask?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a Django & Flask component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

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D. Randomly change configuration

4. Which design goal is most important for a production Django & Flask solution?

A. Reliability, security, maintainability, and observability

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C. No documentation

D. Single points of failure

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A. Consistency and repeatability

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7. Which practice is generally recommended when working with Django & Flask?

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D. Randomly change configuration

9. Which design goal is most important for a production Django & Flask solution in a troubleshooting context?

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10. What is a common reason to automate Django & Flask operations?

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Beginner Level Answer Key & Explanations

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Intermediate Level (10 Questions)

1. In Django & Flask, which statement best describes a core concept?

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Advanced Level (10 Questions)

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6. FastAPI

Category: Backend **Exam:** 30 Questions

Beginner Level (10 Questions)

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B. A filesystem permission unrelated to the technology

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Advanced Level (10 Questions)

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7. NestJS

Category: Backend **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In NestJS, which statement best describes a core concept?

A. The primary mechanism used by NestJS

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with NestJS?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a NestJS component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

4. Which design goal is most important for a production NestJS solution?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

5. What is a common reason to automate NestJS operations?

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D. Single points of failure

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Intermediate Level (10 Questions)

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B. Maximum complexity

C. No documentation

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Intermediate Level Answer Key & Explanations

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8. Node.js

Category: Backend **Exam:** 30 Questions

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9. Ruby on Rails

Category: Backend **Exam:** 30 Questions

Beginner Level (10 Questions)

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10. Spring Boot

Category: Backend **Exam:** 30 Questions

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8. A production issue occurs in a Spring Boot component. What should you check first? Choose the most appropriate option.

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

9. Which design goal is most important for a production Spring Boot solution in a troubleshooting context?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

10. What is a common reason to automate Spring Boot operations?

A. Consistency and repeatability

B. To eliminate all testing

C. To hide failures

D. To prevent versioning

Advanced Level Answer Key & Explanations

1. **A** — The first option represents a central concept used in Spring Boot. The other options are distractors unrelated to the core mechanism.
2. **A** — Versioning, testing, least privilege, and observability are broadly applicable engineering practices.

3. A — Start with evidence and recent changes before making disruptive modifications.

4. A — Production systems should balance reliability, security, maintainability, and observability.

5. A — Automation reduces manual variation and makes repeatable operations easier to test and audit.

6. A — The first option represents a central concept used in Spring Boot. The other options are distractors unrelated to the core mechanism.

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11. AWS

Category: Cloud **Exam:** 30 Questions

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13. GCP

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14. Apache Airflow

Category: Data Engineering **Exam:** 30 Questions

Beginner Level (10 Questions)

1. What does DAG stand for in Airflow?

A. Directed Acyclic Graph

B. Data Access Gateway

C. Distributed API Group

D. Dynamic Agent Grid

2. What is an Airflow task?

A. A unit of executable work

B. A database table

C. A scheduler node only

D. A log file

3. What schedules DAG runs?

A. The Airflow scheduler

B. The web server only

C. A database trigger only

D. Git

4. What is XCom used for?

A. Small pieces of data exchanged between tasks

B. Large data lake storage

C. Container images

D. Network routing

5. Why should tasks be idempotent?

A. Retries should not cause unintended duplicate effects

B. It makes DAGs cyclic

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Beginner Level Answer Key & Explanations

1. A — A DAG defines tasks and dependencies without cycles.

2. A — Tasks are the executable units that form a DAG.

3. A — The scheduler determines when tasks and DAG runs should be created.

4. A — XCom is intended for small inter-task metadata or values, not bulk datasets.

5. A — Idempotent tasks behave safely when Airflow retries them.

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10. A — Idempotent tasks behave safely when Airflow retries them.

15. Apache Kafka

Category: Data Engineering **Exam:** 30 Questions

Beginner Level (10 Questions)

1. What is a Kafka topic?

A. A named stream of records

B. A database server

C. A consumer process

D. A firewall rule

2. What is a partition?

A. An ordered log within a topic

B. A user account

C. A broker password

D. A schema registry only

3. What is a consumer group?

A. Consumers cooperating to process topic partitions

B. A broker cluster name only

C. A topic alias

D. A security policy

4. What is an offset?

A. A record position within a partition

B. A network port

C. A schema version only

D. A topic name

5. Why are keys useful in Kafka?

A. They can determine partition placement

B. They encrypt payloads

C. They create brokers

D. They disable retention

6. What is a Kafka topic is the best choice?

A. A named stream of records

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Beginner Level Answer Key & Explanations

1. A — Topics organize records into named streams.

2. A — Partitions provide ordered records and enable parallelism.

3. A — A consumer group distributes partitions among its members.

4. A — Offsets identify record positions and support consumption progress tracking.

5. A — The key can be hashed to choose a partition, preserving per-key ordering.

6. A — Topics organize records into named streams.

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16. Apache Spark

Category: Data Engineering **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In Apache Spark, which statement best describes a core concept?

A. The primary mechanism used by Apache Spark

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with Apache Spark?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a Apache Spark component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

4. Which design goal is most important for a production Apache Spark solution?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

5. What is a common reason to automate Apache Spark operations?

A. Consistency and repeatability

B. To eliminate all testing

C. To hide failures

D. To prevent versioning

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D. To prevent versioning

Beginner Level Answer Key & Explanations

1. A — The first option represents a central concept used in Apache Spark. The other options are distractors unrelated to the core mechanism.

2. A — Versioning, testing, least privilege, and observability are broadly applicable engineering practices.

3. A — Start with evidence and recent changes before making disruptive modifications.

4. A — Production systems should balance reliability, security, maintainability, and observability.

5. A — Automation reduces manual variation and makes repeatable operations easier to test and audit.

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17. Snowflake & Data Warehousing

Category: Data Engineering **Exam:** 30 Questions

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18. Tableau & Power BI

Category: Data Engineering **Exam:** 30 Questions

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B. To eliminate all testing

C. To hide failures

D. To prevent versioning

6. In Tableau & Power BI, which statement best describes a core concept is the best choice?

A. The primary mechanism used by Tableau & Power BI

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

7. Which practice is generally recommended when working with Tableau & Power BI?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

8. A production issue occurs in a Tableau & Power BI component. What should you check first? Choose the most appropriate option.

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

9. Which design goal is most important for a production Tableau & Power BI solution in a troubleshooting context?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

10. What is a common reason to automate Tableau & Power BI operations?

A. Consistency and repeatability

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Intermediate Level Answer Key & Explanations

1. A — The first option represents a central concept used in Tableau & Power BI. The other options are distractors unrelated to the core mechanism.

2. A — Versioning, testing, least privilege, and observability are broadly applicable engineering practices.

3. A — Start with evidence and recent changes before making disruptive modifications.

4. A — Production systems should balance reliability, security, maintainability, and observability.

5. A — Automation reduces manual variation and makes repeatable operations easier to test and audit.

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Advanced Level (10 Questions)

1. In Tableau & Power BI, which statement best describes a core concept?

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19. MongoDB & NoSQL

Category: Databases **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In MongoDB & NoSQL, which statement best describes a core concept?

A. The primary mechanism used by MongoDB & NoSQL

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with MongoDB & NoSQL?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a MongoDB & NoSQL component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

4. Which design goal is most important for a production MongoDB & NoSQL solution?

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B. Maximum complexity

C. No documentation

D. Single points of failure

5. What is a common reason to automate MongoDB & NoSQL operations?

A. Consistency and repeatability

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D. To prevent versioning

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D. A deprecated convention

7. Which practice is generally recommended when working with MongoDB & NoSQL?

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C. Hard-code every environment value

D. Avoid monitoring

8. A production issue occurs in a MongoDB & NoSQL component. What should you check first? Choose the most appropriate option.

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

9. Which design goal is most important for a production MongoDB & NoSQL solution in a troubleshooting context?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

10. What is a common reason to automate MongoDB & NoSQL operations?

A. Consistency and repeatability

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D. To prevent versioning

Beginner Level Answer Key & Explanations

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20. MySQL

Category: Databases **Exam:** 30 Questions

Beginner Level (10 Questions)

1. Which SQL clause filters rows before grouping?

A. WHERE

B. HAVING

C. GROUP BY

D. ORDER BY

2. Which clause filters grouped results?

A. HAVING

B. WHERE

C. FILTER BY

D. GROUP

3. Which command adds new rows to a table?

A. INSERT

B. ADD

C. APPEND

D. CREATE

4. Which operation combines rows from related tables?

A. JOIN

B. UNION ONLY

C. MERGE ONLY

D. LINK

5. Which index structure is commonly used to accelerate equality and range lookups?

A. B-tree

B. Heap only

C. Stack

D. Bitmap only

6. In a production environment, which SQL clause filters rows before grouping is the best choice?

A. WHERE

B. HAVING

C. GROUP BY

D. ORDER BY

7. Which clause filters grouped results?

A. HAVING

B. WHERE

C. FILTER BY

D. GROUP

8. Which command adds new rows to a table? Choose the most appropriate option.

A. INSERT

B. ADD

C. APPEND

D. CREATE

9. Which operation combines rows from related tables in a troubleshooting context?

A. JOIN

B. UNION ONLY

C. MERGE ONLY

D. LINK

10. For a modern deployment, which index structure is commonly used to accelerate equality and range lookups?

A. B-tree

B. Heap only

C. Stack

D. Bitmap only

Beginner Level Answer Key & Explanations

1. A — `WHERE` filters individual rows before grouping and aggregation.

2. A — `HAVING` applies conditions after grouping and aggregation.

3. A — `INSERT` adds rows to an existing table.

4. A — A `JOIN` combines rows using related columns or join conditions.

5. A — B-tree indexes efficiently support many equality and range predicates.

6. A — `WHERE` filters individual rows before grouping and aggregation.

7. A — `HAVING` applies conditions after grouping and aggregation.

8. A — `INSERT` adds rows to an existing table.

9. A — A `JOIN` combines rows using related columns or join conditions.

10. A — B-tree indexes efficiently support many equality and range predicates.

Intermediate Level (10 Questions)

1. Which SQL clause filters rows before grouping?

A. WHERE

B. HAVING

C. GROUP BY

D. ORDER BY

2. Which clause filters grouped results?

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Intermediate Level Answer Key & Explanations

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Advanced Level (10 Questions)

1. Which SQL clause filters rows before grouping?

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Advanced Level Answer Key & Explanations

1. A — `WHERE` filters individual rows before grouping and aggregation.

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8. A — `INSERT` adds rows to an existing table.

9. A — A `JOIN` combines rows using related columns or join conditions.

10. A — B-tree indexes efficiently support many equality and range predicates.

21. Oracle Database

Category: Databases **Exam:** 30 Questions

Beginner Level (10 Questions)

1. Which SQL clause filters rows before grouping?

A. WHERE

B. HAVING

C. GROUP BY

D. ORDER BY

2. Which clause filters grouped results?

A. HAVING

B. WHERE

C. FILTER BY

D. GROUP

3. Which command adds new rows to a table?

A. INSERT

B. ADD

C. APPEND

D. CREATE

4. Which operation combines rows from related tables?

A. JOIN

B. UNION ONLY

C. MERGE ONLY

D. LINK

5. Which index structure is commonly used to accelerate equality and range lookups?

A. B-tree

B. Heap only

C. Stack

D. Bitmap only

6. In a production environment, which SQL clause filters rows before grouping is the best choice?

A. WHERE

B. HAVING

C. GROUP BY

D. ORDER BY

7. Which clause filters grouped results?

A. HAVING

B. WHERE

C. FILTER BY

D. GROUP

8. Which command adds new rows to a table? Choose the most appropriate option.

A. INSERT

B. ADD

C. APPEND

D. CREATE

9. Which operation combines rows from related tables in a troubleshooting context?

A. JOIN

B. UNION ONLY

C. MERGE ONLY

D. LINK

10. For a modern deployment, which index structure is commonly used to accelerate equality and range lookups?

A. B-tree

B. Heap only

C. Stack

D. Bitmap only

Beginner Level Answer Key & Explanations

1. **A** — `WHERE` filters individual rows before grouping and aggregation.
2. **A** — `HAVING` applies conditions after grouping and aggregation.
3. **A** — `INSERT` adds rows to an existing table.
4. **A** — A `JOIN` combines rows using related columns or join conditions.
5. **A** — B-tree indexes efficiently support many equality and range predicates.
6. **A** — `WHERE` filters individual rows before grouping and aggregation.
7. **A** — `HAVING` applies conditions after grouping and aggregation.
8. **A** — `INSERT` adds rows to an existing table.
9. **A** — A `JOIN` combines rows using related columns or join conditions.
10. **A** — B-tree indexes efficiently support many equality and range predicates.

Intermediate Level (10 Questions)

1. Which SQL clause filters rows before grouping?

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A. JOIN

B. UNION ONLY

C. MERGE ONLY

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B. Heap only

C. Stack

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Advanced Level (10 Questions)

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Advanced Level Answer Key & Explanations

1. A — `WHERE` filters individual rows before grouping and aggregation.

2. A — `HAVING` applies conditions after grouping and aggregation.

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7. A — `HAVING` applies conditions after grouping and aggregation.

8. A — `INSERT` adds rows to an existing table.

9. A — A `JOIN` combines rows using related columns or join conditions.

10. A — B-tree indexes efficiently support many equality and range predicates.

22. PostgreSQL

Category: Databases **Exam:** 30 Questions

Beginner Level (10 Questions)

1. Which SQL clause filters rows before grouping?

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2. Which clause filters grouped results?

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C. Stack

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Intermediate Level (10 Questions)

1. Which SQL clause filters rows before grouping?

A. WHERE

B. HAVING

C. GROUP BY

D. ORDER BY

2. Which clause filters grouped results?

A. HAVING

B. WHERE

C. FILTER BY

D. GROUP

3. Which command adds new rows to a table?

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B. ADD

C. APPEND

D. CREATE

4. Which operation combines rows from related tables?

A. JOIN

B. UNION ONLY

C. MERGE ONLY

D. LINK

5. Which index structure is commonly used to accelerate equality and range lookups?

A. B-tree

B. Heap only

C. Stack

D. Bitmap only

6. In a production environment, which SQL clause filters rows before grouping is the best choice?

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23. Redis & Caching

Category: Databases **Exam:** 30 Questions

Beginner Level (10 Questions)

1. What is Redis commonly used for?

A. In-memory caching and data structures

B. Compiling Java

C. Container orchestration

D. DNS registration

2. What does TTL mean in Redis?

A. Time to live

B. Transaction table lock

C. Total transfer limit

D. Thread timeout list

3. Which Redis structure stores field-value pairs under one key?

A. Hash

B. Stack only

C. Binary tree

D. Graph

4. Why use cache-aside?

A. The application loads missing data into cache on demand

B. The cache writes every DB change automatically

C. It removes the database

D. It disables expiration

5. What is cache stampede?

A. Many requests simultaneously recompute an expired value

B. A Redis syntax error

C. A broker failure

D. A database index

6. What is Redis commonly used for is the best choice?

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Beginner Level Answer Key & Explanations

1. A — Redis is an in-memory data store often used for caching, queues, and fast lookups.

2. A — TTL specifies how long a key remains before expiration.

3. A — A Redis hash stores multiple field-value pairs associated with a key.

4. A — Cache-aside lets the application read the database on a miss and populate the cache.

5. A — A stampede occurs when many clients miss the same cache entry and overload the backend.

6. A — Redis is an in-memory data store often used for caching, queues, and fast lookups.

7. A — TTL specifies how long a key remains before expiration.

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24. SQL

Category: Databases **Exam:** 30 Questions

Beginner Level (10 Questions)

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C. GROUP BY

D. ORDER BY

2. Which clause filters grouped results?

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3. Which command adds new rows to a table?

A. INSERT

B. ADD

C. APPEND

D. CREATE

4. Which operation combines rows from related tables?

A. JOIN

B. UNION ONLY

C. MERGE ONLY

D. LINK

5. Which index structure is commonly used to accelerate equality and range lookups?

A. B-tree

B. Heap only

C. Stack

D. Bitmap only

6. In a production environment, which SQL clause filters rows before grouping is the best choice?

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25. Ansible

Category: DevOps **Exam:** 30 Questions

Beginner Level (10 Questions)

1. What format is commonly used for Ansible playbooks?

A. YAML

B. XML

C. CSV

D. INI only

2. What is an Ansible inventory?

A. A definition of managed hosts and groups

B. A task log only

C. A Git repository

D. A package registry

3. Which Ansible module is commonly used to run a command?

A. command

B. execute

C. shellcmd

D. run

4. What does idempotent automation mean?

A. Repeated execution converges to the same desired state

B. Every run changes the server

C. Tasks never fail

D. Only one host can be managed

5. What is a role used for?

A. Packaging reusable Ansible automation

B. Creating a Linux user only

C. Replacing inventory

D. Encrypting network traffic

6. What format is commonly used for Ansible playbooks is the best choice?

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Beginner Level Answer Key & Explanations

1. A — Ansible playbooks are typically written in YAML.

2. A — The inventory identifies hosts Ansible can manage and group them logically.

3. A — The `command` module executes a command without shell interpretation.

4. A — Idempotent tasks can be safely repeated without unwanted cumulative changes.

5. A — Roles organize variables, tasks, handlers, templates, and files into reusable units.

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26. CI/CD & DevOps

Category: DevOps **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In CI/CD & DevOps, which statement best describes a core concept?

A. The primary mechanism used by CI/CD & DevOps

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with CI/CD & DevOps?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a CI/CD & DevOps component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

4. Which design goal is most important for a production CI/CD & DevOps solution?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

5. What is a common reason to automate CI/CD & DevOps operations?

A. Consistency and repeatability

B. To eliminate all testing

C. To hide failures

D. To prevent versioning

6. In CI/CD & DevOps, which statement best describes a core concept is the best choice?

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C. No documentation

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A. Consistency and repeatability

B. To eliminate all testing

C. To hide failures

D. To prevent versioning

Beginner Level Answer Key & Explanations

1. A — The first option represents a central concept used in CI/CD & DevOps. The other options are distractors unrelated to the core mechanism.

2. A — Versioning, testing, least privilege, and observability are broadly applicable engineering practices.

3. A — Start with evidence and recent changes before making disruptive modifications.

4. A — Production systems should balance reliability, security, maintainability, and observability.

5. A — Automation reduces manual variation and makes repeatable operations easier to test and audit.

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3. A production issue occurs in a CI/CD & DevOps component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

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D. Randomly change configuration

4. Which design goal is most important for a production CI/CD & DevOps solution?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

5. What is a common reason to automate CI/CD & DevOps operations?

A. Consistency and repeatability

B. To eliminate all testing

C. To hide failures

D. To prevent versioning

6. In CI/CD & DevOps, which statement best describes a core concept is the best choice?

A. The primary mechanism used by CI/CD & DevOps

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

7. Which practice is generally recommended when working with CI/CD & DevOps?

A. Use version control, testing, and least privilege

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Intermediate Level Answer Key & Explanations

1. A — The first option represents a central concept used in CI/CD & DevOps. The other options are distractors unrelated to the core mechanism.

2. A — Versioning, testing, least privilege, and observability are broadly applicable engineering practices.

3. A — Start with evidence and recent changes before making disruptive modifications.

4. A — Production systems should balance reliability, security, maintainability, and observability.

5. A — Automation reduces manual variation and makes repeatable operations easier to test and audit.

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27. Docker & Kubernetes

Category: DevOps **Exam:** 30 Questions

Beginner Level (10 Questions)

1. What is a container image?

A. An immutable template used to create containers

B. A running VM

C. A Kubernetes node

D. A network cable

2. Which Dockerfile instruction specifies the base image?

A. FROM

B. BASE

C. IMAGE

D. START

3. What does a Kubernetes Deployment primarily manage?

A. Replica sets and desired application state

B. DNS zones only

C. Physical switches

D. Git branches

4. Which Kubernetes object exposes an application through a stable network endpoint?

A. Service

B. ConfigMap

C. Secret

D. Job

5. What is a readiness probe used for?

A. Determining whether a container is ready to receive traffic

B. Restarting every container

C. Building an image

D. Encrypting secrets

6. What is a container image is the best choice?

A. An immutable template used to create containers

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Beginner Level Answer Key & Explanations

1. A — An image packages filesystem layers and metadata used to create containers.

2. A — `FROM` selects the base image for a Docker build.

3. A — A Deployment declaratively manages replicated Pods through ReplicaSets.

4. A — A Service provides stable discovery and load balancing for selected Pods.

5. A — Readiness controls whether the Pod should receive traffic.

6. A — An image packages filesystem layers and metadata used to create containers.

7. A — `FROM` selects the base image for a Docker build.

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28. Git

Category: DevOps **Exam:** 30 Questions

Beginner Level (10 Questions)

1. What command creates a new Git repository in the current directory?

A. git init

B. git start

C. git create

D. git repo

2. Which command records staged changes in repository history?

A. git commit

B. git save

C. git record

D. git snapshot

3. Which command shows the current working-tree and staging-area state?

A. git status

B. git state

C. git check

D. git info

4. Which command creates and switches to a new branch?

A. git switch -c feature

B. git branch --new feature

C. git branch feature --switch

D. git new feature

5. Which command is designed to undo a published commit without rewriting history?

A. git revert

B. git erase

C. git reset --hard

D. git drop

6. What command creates a new Git repository in the current directory is the best choice?

A. git init

B. git start

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10. For a modern deployment, which command is designed to undo a published commit without rewriting history?

A. git revert

B. git erase

C. git reset --hard

D. git drop

Beginner Level Answer Key & Explanations

1. A — `git init` initializes a new repository by creating the `.git` directory.

2. A — `git commit` creates a new commit from the staged snapshot.

3. A — `git status` reports modified, staged, and untracked files.

4. A — `git switch -c` creates the branch and checks it out.

5. A — `git revert` creates a new commit that reverses the selected commit.

6. A — `git init` initializes a new repository by creating the `.git` directory.

7. A — `git commit` creates a new commit from the staged snapshot.

8. A — `git status` reports modified, staged, and untracked files.

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29. Terraform

Category: DevOps **Exam:** 30 Questions

Beginner Level (10 Questions)

1. What language is used for Terraform configuration?

A. HCL

B. YAML only

C. SQL

D. Lua only

2. What is Terraform state used for?

A. Tracking managed resources and their attributes

B. Storing passwords by default

C. Replacing Git

D. Running containers

3. What does `terraform plan` do?

A. Shows proposed changes

B. Applies all changes

C. Deletes state

D. Creates a Git branch

4. What does `terraform apply` do?

A. Applies the planned configuration changes

B. Only validates syntax

C. Removes providers

D. Formats files

5. Why use remote state with locking?

A. To coordinate shared state and prevent concurrent updates

B. To compile HCL

C. To replace IAM

D. To disable drift

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Beginner Level Answer Key & Explanations

1. A — Terraform commonly uses HashiCorp Configuration Language (HCL).

2. A — State maps configuration resources to real infrastructure objects.

3. A — `plan` calculates and displays the intended infrastructure changes.

4. A — `apply` executes the proposed changes against configured providers.

5. A — Remote state centralizes state and locking helps prevent conflicting operations.

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30. Jira & Agile

Category: Enterprise **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In Jira & Agile, which statement best describes a core concept?

A. The primary mechanism used by Jira & Agile

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with Jira & Agile?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a Jira & Agile component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

4. Which design goal is most important for a production Jira & Agile solution?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

5. What is a common reason to automate Jira & Agile operations?

A. Consistency and repeatability

B. To eliminate all testing

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D. To prevent versioning

Beginner Level Answer Key & Explanations

1. A — The first option represents a central concept used in Jira & Agile. The other options are distractors unrelated to the core mechanism.

2. A — Versioning, testing, least privilege, and observability are broadly applicable engineering practices.

3. A — Start with evidence and recent changes before making disruptive modifications.

4. A — Production systems should balance reliability, security, maintainability, and observability.

5. A — Automation reduces manual variation and makes repeatable operations easier to test and audit.

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10. A — Automation reduces manual variation and makes repeatable operations easier to test and audit.

Intermediate Level (10 Questions)

1. In Jira & Agile, which statement best describes a core concept?

A. The primary mechanism used by Jira & Agile

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with Jira & Agile?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a Jira & Agile component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

4. Which design goal is most important for a production Jira & Agile solution?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

5. What is a common reason to automate Jira & Agile operations?

A. Consistency and repeatability

B. To eliminate all testing

C. To hide failures

D. To prevent versioning

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B. Maximum complexity

C. No documentation

D. Single points of failure

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C. To hide failures

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31. SAP

Category: Enterprise **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In SAP, which statement best describes a core concept?

A. The primary mechanism used by SAP

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with SAP?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a SAP component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

4. Which design goal is most important for a production SAP solution?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

5. What is a common reason to automate SAP operations?

A. Consistency and repeatability

B. To eliminate all testing

C. To hide failures

D. To prevent versioning

6. In SAP, which statement best describes a core concept is the best choice?

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B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

8. A production issue occurs in a SAP component. What should you check first? Choose the most appropriate option.

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

9. Which design goal is most important for a production SAP solution in a troubleshooting context?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

10. What is a common reason to automate SAP operations?

A. Consistency and repeatability

B. To eliminate all testing

C. To hide failures

D. To prevent versioning

Beginner Level Answer Key & Explanations

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Intermediate Level (10 Questions)

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Advanced Level (10 Questions)

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32. React & Next.js

Category: Frontend **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In React & Next.js, which statement best describes a core concept?

A. The primary mechanism used by React & Next.js

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with React & Next.js?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a React & Next.js component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

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4. Which design goal is most important for a production React & Next.js solution?

A. Reliability, security, maintainability, and observability

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Beginner Level Answer Key & Explanations

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Intermediate Level (10 Questions)

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Advanced Level (10 Questions)

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33. Vue.js & Angular

Category: Frontend **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In Vue.js & Angular, which statement best describes a core concept?

A. The primary mechanism used by Vue.js & Angular

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with Vue.js & Angular?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a Vue.js & Angular component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

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4. Which design goal is most important for a production Vue.js & Angular solution?

A. Reliability, security, maintainability, and observability

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5. What is a common reason to automate Vue.js & Angular operations?

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Intermediate Level (10 Questions)

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A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

9. Which design goal is most important for a production Vue.js & Angular solution in a troubleshooting context?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

10. What is a common reason to automate Vue.js & Angular operations?

A. Consistency and repeatability

B. To eliminate all testing

C. To hide failures

D. To prevent versioning

Intermediate Level Answer Key & Explanations

1. A — The first option represents a central concept used in Vue.js & Angular. The other options are distractors unrelated to the core mechanism.

2. A — Versioning, testing, least privilege, and observability are broadly applicable engineering practices.

3. A — Start with evidence and recent changes before making disruptive modifications.

4. A — Production systems should balance reliability, security, maintainability, and observability.

5. A — Automation reduces manual variation and makes repeatable operations easier to test and audit.

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Advanced Level (10 Questions)

1. In Vue.js & Angular, which statement best describes a core concept?

A. The primary mechanism used by Vue.js & Angular

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with Vue.js & Angular?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

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34. Linux

Category: Infrastructure **Exam:** 30 Questions

Beginner Level (10 Questions)

1. Which command lists directory contents?

A. ls

B. dirlist

C. show

D. files

2. Which command prints the current working directory?

A. pwd

B. cwd

C. where

D. path

3. Which command changes directory?

A. cd

B. chdirx

C. move

D. goto

4. Which operator redirects standard output to a file, replacing its contents?

A. >

B. >>

C. <

D. |

5. Which command changes file permissions?

A. chmod

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C. perm

D. setmode

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10. For a modern deployment, which command changes file permissions?

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D. setmode

Beginner Level Answer Key & Explanations

1. A — `ls` lists files and directories.

2. A — `pwd` prints the current directory path.

3. A — `cd` changes the shell's current working directory.

4. A — `>` redirects stdout and truncates/replaces the destination file.

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35. Networking

Category: Infrastructure **Exam:** 30 Questions

Beginner Level (10 Questions)

1. Which protocol provides reliable connection-oriented transport?

A. TCP

B. UDP

C. ARP

D. ICMP

2. Which protocol maps IPv4 addresses to MAC addresses on a local network?

A. ARP

B. DNS

C. DHCP

D. HTTP

3. What does DNS primarily do?

A. Maps names to network records such as IP addresses

B. Encrypts traffic

C. Assigns MAC addresses

D. Routes packets

4. What does a subnet mask determine?

A. The network and host portions of an IP address

B. The TCP port

C. The DNS server software

D. The encryption key

5. What is NAT commonly used for?

A. Translating addresses between network namespaces

B. Encrypting HTTP

C. Replacing DNS

D. Creating VLAN tags only

6. In a production environment, which protocol provides reliable connection-oriented transport is the best choice?

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Beginner Level Answer Key & Explanations

1. A — TCP provides reliable, ordered byte-stream delivery.

2. A — ARP resolves IPv4 addresses to link-layer MAC addresses.

3. A — DNS resolves domain names and other names to resource records.

4. A — The mask defines which address bits identify the network versus host.

5. A — NAT translates source or destination addresses, often between private and public networks.

6. A — TCP provides reliable, ordered byte-stream delivery.

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36. PowerShell

Category: Infrastructure **Exam:** 30 Questions

Beginner Level (10 Questions)

1. Which command lists directory contents?

A. ls

B. dirlist

C. show

D. files

2. Which command prints the current working directory?

A. pwd

B. cwd

C. where

D. path

3. Which command changes directory?

A. cd

B. chdirx

C. move

D. goto

4. Which operator redirects standard output to a file, replacing its contents?

A. >

B. >>

C. <

D. |

5. Which command changes file permissions?

A. chmod

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Beginner Level Answer Key & Explanations

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37. VMware & Virtualisation

Category: Infrastructure **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In VMware & Virtualisation, which statement best describes a core concept?

A. The primary mechanism used by VMware & Virtualisation

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with VMware & Virtualisation?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a VMware & Virtualisation component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

4. Which design goal is most important for a production VMware & Virtualisation solution?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

5. What is a common reason to automate VMware & Virtualisation operations?

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6. A — The first option represents a central concept used in VMware & Virtualisation. The other options are distractors unrelated to the core mechanism.

7. A — Versioning, testing, least privilege, and observability are broadly applicable engineering practices.

8. A — Start with evidence and recent changes before making disruptive modifications.

9. A — Production systems should balance reliability, security, maintainability, and observability.

10. A — Automation reduces manual variation and makes repeatable operations easier to test and audit.

Advanced Level (10 Questions)

1. In VMware & Virtualisation, which statement best describes a core concept?

A. The primary mechanism used by VMware & Virtualisation

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with VMware & Virtualisation?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a VMware & Virtualisation component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

4. Which design goal is most important for a production VMware & Virtualisation solution?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

5. What is a common reason to automate VMware & Virtualisation operations?

A. Consistency and repeatability

B. To eliminate all testing

C. To hide failures

D. To prevent versioning

6. In VMware & Virtualisation, which statement best describes a core concept is the best choice?

A. The primary mechanism used by VMware & Virtualisation

B. A filesystem permission unrelated to the technology

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7. Which practice is generally recommended when working with VMware & Virtualisation?

A. Use version control, testing, and least privilege

B. Disable all logging

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8. A production issue occurs in a VMware & Virtualisation component. What should you check first? Choose the most appropriate option.

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

9. Which design goal is most important for a production VMware & Virtualisation solution in a troubleshooting context?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

10. What is a common reason to automate VMware & Virtualisation operations?

A. Consistency and repeatability

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Advanced Level Answer Key & Explanations

1. A — The first option represents a central concept used in VMware & Virtualisation. The other options are distractors unrelated to the core mechanism.

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3. A — Start with evidence and recent changes before making disruptive modifications.

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10. A — Automation reduces manual variation and makes repeatable operations easier to test and audit.

38. Bash & Shell Scripting

Category: Languages **Exam:** 30 Questions

Beginner Level (10 Questions)

1. Which command lists directory contents?

A. ls

B. dirlist

C. show

D. files

2. Which command prints the current working directory?

A. pwd

B. cwd

C. where

D. path

3. Which command changes directory?

A. cd

B. chdirx

C. move

D. goto

4. Which operator redirects standard output to a file, replacing its contents?

A. >

B. >>

C. <

D. |

5. Which command changes file permissions?

A. chmod

B. chperm

C. perm

D. setmode

6. In a production environment, which command lists directory contents is the best choice?

A. ls

B. dirlist

C. show

D. files

7. Which command prints the current working directory?

A. pwd

B. cwd

C. where

D. path

8. Which command changes directory? Choose the most appropriate option.

A. cd

B. chdirx

C. move

D. goto

9. Which operator redirects standard output to a file, replacing its contents in a troubleshooting context?

A. >

B. >>

C. <

D. |

10. For a modern deployment, which command changes file permissions?

A. chmod

B. chperm

C. perm

D. setmode

Beginner Level Answer Key & Explanations

1. A — `ls` lists files and directories.

2. A — `pwd` prints the current directory path.

3. A — `cd` changes the shell's current working directory.

4. A — `>` redirects stdout and truncates/replaces the destination file.

5. A — `chmod` changes Unix permission bits.

6. A — `ls` lists files and directories.

7. A — `pwd` prints the current directory path.

8. A — `cd` changes the shell's current working directory.

9. A — `>` redirects stdout and truncates/replaces the destination file.

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Intermediate Level (10 Questions)

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Intermediate Level Answer Key & Explanations

1. A — `ls` lists files and directories.

2. A — `pwd` prints the current directory path.

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Advanced Level (10 Questions)

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39. C# & .NET

Category: Languages **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In C# & .NET, which statement best describes a core concept?

A. The primary mechanism used by C# & .NET

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with C# & .NET?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a C# & .NET component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

4. Which design goal is most important for a production C# & .NET solution?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

5. What is a common reason to automate C# & .NET operations?

A. Consistency and repeatability

B. To eliminate all testing

C. To hide failures

D. To prevent versioning

6. In C# & .NET, which statement best describes a core concept is the best choice?

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D. Single points of failure

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A. Consistency and repeatability

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D. To prevent versioning

Beginner Level Answer Key & Explanations

1. A — The first option represents a central concept used in C# & .NET. The other options are distractors unrelated to the core mechanism.

2. A — Versioning, testing, least privilege, and observability are broadly applicable engineering practices.

3. A — Start with evidence and recent changes before making disruptive modifications.

4. A — Production systems should balance reliability, security, maintainability, and observability.

5. A — Automation reduces manual variation and makes repeatable operations easier to test and audit.

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40. Go (Golang)

Category: Languages **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In Go (Golang), which statement best describes a core concept?

A. The primary mechanism used by Go (Golang)

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with Go (Golang)?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a Go (Golang) component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

4. Which design goal is most important for a production Go (Golang) solution?

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C. No documentation

D. Single points of failure

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Beginner Level Answer Key & Explanations

1. **A** — The first option represents a central concept used in Go (Golang). The other options are distractors unrelated to the core mechanism.
2. **A** — Versioning, testing, least privilege, and observability are broadly applicable engineering practices.
3. **A** — Start with evidence and recent changes before making disruptive modifications.
4. **A** — Production systems should balance reliability, security, maintainability, and observability.
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Intermediate Level (10 Questions)

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C. A hardware-only feature

D. A deprecated convention

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B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a Go (Golang) component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

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B. Maximum complexity

C. No documentation

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B. Maximum complexity

C. No documentation

D. Single points of failure

10. What is a common reason to automate Go (Golang) operations?

A. Consistency and repeatability

B. To eliminate all testing

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Advanced Level Answer Key & Explanations

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8. A — Start with evidence and recent changes before making disruptive modifications.

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10. A — Automation reduces manual variation and makes repeatable operations easier to test and audit.

41. Java

Category: Languages **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In Java, which statement best describes a core concept?

A. The primary mechanism used by Java

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with Java?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a Java component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

4. Which design goal is most important for a production Java solution?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

5. What is a common reason to automate Java operations?

A. Consistency and repeatability

B. To eliminate all testing

C. To hide failures

D. To prevent versioning

6. In Java, which statement best describes a core concept is the best choice?

A. The primary mechanism used by Java

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C. Hard-code every environment value

D. Avoid monitoring

8. A production issue occurs in a Java component. What should you check first? Choose the most appropriate option.

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

9. Which design goal is most important for a production Java solution in a troubleshooting context?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

10. What is a common reason to automate Java operations?

A. Consistency and repeatability

B. To eliminate all testing

C. To hide failures

D. To prevent versioning

Beginner Level Answer Key & Explanations

1. **A** — The first option represents a central concept used in Java. The other options are distractors unrelated to the core mechanism.
2. **A** — Versioning, testing, least privilege, and observability are broadly applicable engineering practices.
3. **A** — Start with evidence and recent changes before making disruptive modifications.
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Intermediate Level (10 Questions)

1. In Java, which statement best describes a core concept?

A. The primary mechanism used by Java

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with Java?

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42. JavaScript

Category: Languages **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In JavaScript, which statement best describes a core concept?

A. The primary mechanism used by JavaScript

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C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with JavaScript?

A. Use version control, testing, and least privilege

B. Disable all logging

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3. A production issue occurs in a JavaScript component. What should you check first?

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43. PHP & Laravel

Category: Languages **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In PHP & Laravel, which statement best describes a core concept?

A. The primary mechanism used by PHP & Laravel

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with PHP & Laravel?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a PHP & Laravel component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

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D. Randomly change configuration

4. Which design goal is most important for a production PHP & Laravel solution?

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Beginner Level Answer Key & Explanations

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44. Python

Category: Languages **Exam:** 30 Questions

Beginner Level (10 Questions)

1. Which keyword defines a function in Python?

A. def

B. func

C. function

D. define

2. Which data type is mutable?

A. list

B. tuple

C. str

D. frozenset

3. What does `len()` return for a sequence?

A. Number of elements

B. Memory size

C. Last index

D. Data type

4. Which construct handles exceptions?

A. try/except

B. catch/throw

C. handle/error

D. safe/rescue

5. What does `if __name__ == '__main__':` commonly do?

A. Runs code only when the file is executed directly

B. Imports the file automatically

C. Creates a package

D. Enables debugging

6. In a production environment, which keyword defines a function in Python is the best choice?

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B. func

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Beginner Level Answer Key & Explanations

1. A — Python uses `def` to declare functions.

2. A — Python lists can be changed after creation.

3. A — `len()` returns the number of items in a sequence or collection.

4. A — Python uses `try` with `except` clauses for exception handling.

5. A — The condition is true when the module is the program's entry script.

6. A — Python uses `def` to declare functions.

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Intermediate Level (10 Questions)

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Intermediate Level Answer Key & Explanations

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3. **A** — `len()` returns the number of items in a sequence or collection.
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Advanced Level (10 Questions)

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C. function

D. define

2. Which data type is mutable?

A. list

B. tuple

C. str

D. frozenset

3. What does `len()` return for a sequence?

A. Number of elements

B. Memory size

C. Last index

D. Data type

4. Which construct handles exceptions?

A. try/except

B. catch/throw

C. handle/error

D. safe/rescue

5. What does `if __name__ == '__main__':` commonly do?

A. Runs code only when the file is executed directly

B. Imports the file automatically

C. Creates a package

D. Enables debugging

6. In a production environment, which keyword defines a function in Python is the best choice?

A. def

B. func

C. function

D. define

7. Which data type is mutable?

A. list

B. tuple

C. str

D. frozenset

8. What does `len()` return for a sequence? Choose the most appropriate option.

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B. Memory size

C. Last index

D. Data type

9. Which construct handles exceptions in a troubleshooting context?

A. try/except

B. catch/throw

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D. safe/rescue

10. What does `if __name__ == '__main__':` commonly do?

A. Runs code only when the file is executed directly

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Advanced Level Answer Key & Explanations

1. A — Python uses `def` to declare functions.

2. A — Python lists can be changed after creation.

3. A — `len()` returns the number of items in a sequence or collection.

4. A — Python uses `try` with `except` clauses for exception handling.

5. A — The condition is true when the module is the program's entry script.

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45. Rust

Category: Languages **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In Rust, which statement best describes a core concept?

A. The primary mechanism used by Rust

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with Rust?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a Rust component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

4. Which design goal is most important for a production Rust solution?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

5. What is a common reason to automate Rust operations?

A. Consistency and repeatability

B. To eliminate all testing

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Beginner Level Answer Key & Explanations

1. **A** — The first option represents a central concept used in Rust. The other options are distractors unrelated to the core mechanism.
2. **A** — Versioning, testing, least privilege, and observability are broadly applicable engineering practices.
3. **A** — Start with evidence and recent changes before making disruptive modifications.
4. **A** — Production systems should balance reliability, security, maintainability, and observability.
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46. TypeScript

Category: Languages **Exam:** 30 Questions

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47. RabbitMQ & Messaging

Category: Messaging **Exam:** 30 Questions

Beginner Level (10 Questions)

1. What is a RabbitMQ exchange responsible for?

A. Routing messages to queues

B. Persisting SQL tables

C. Running containers

D. Compiling consumers

2. What is an acknowledgment?

A. A consumer confirms successful processing

B. A broker creates a queue

C. A producer creates an exchange

D. A message is encrypted

3. What does durable mean for a RabbitMQ queue?

A. Its definition survives broker restart

B. It never stores messages

C. It is always temporary

D. It disables acknowledgments

4. What is a dead-letter exchange used for?

A. Routing rejected/expired/dead-lettered messages

B. Encrypting queues

C. Creating users

D. Scheduling cron jobs

5. Why use publisher confirms?

A. To know whether the broker accepted published messages

B. To increase CPU frequency

C. To replace consumers

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Beginner Level Answer Key & Explanations

1. A — Exchanges route published messages to queues according to bindings.

2. A — Acknowledgments tell the broker whether a consumer has accepted a message.

3. A — Durable queues have definitions that survive broker restarts; message durability is a separate concern.

4. A — Dead-lettering provides a route for messages that cannot be normally processed.

5. A — Publisher confirms provide broker-level confirmation of publication.

6. A — Exchanges route published messages to queues according to bindings.

7. A — Acknowledgments tell the broker whether a consumer has accepted a message.

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48. Elasticsearch & ELK Stack

Category: Observability **Exam:** 30 Questions

Beginner Level (10 Questions)

1. What is log ingestion?

A. Collecting and sending log data into a system

B. Deleting logs

C. Compiling code

D. Creating DNS records

2. What is a field in a log event?

A. A named piece of extracted data

B. A physical server

C. A dashboard only

D. A network switch

3. Why parse logs into structured fields?

A. To enable efficient filtering, aggregation, and search

B. To increase file size only

C. To disable alerts

D. To remove timestamps

4. What is an index commonly used for?

A. Organizing searchable data

B. Creating a CPU core

C. Managing Git branches

D. Encrypting passwords

5. What is a correlation search used for?

A. Detecting patterns across events or sources

B. Compressing files only

C. Creating containers

D. Managing DNS

6. What is log ingestion is the best choice?

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D. Managing DNS

Beginner Level Answer Key & Explanations

1. A — Ingestion moves log events from sources into the logging platform.

2. A — Fields represent structured attributes extracted from events.

3. A — Structured fields make analytics and correlation more precise.

4. A — Indexes organize data to support efficient search and retention management.

5. A — Correlation combines multiple signals to identify suspicious or meaningful patterns.

6. A — Ingestion moves log events from sources into the logging platform.

7. A — Fields represent structured attributes extracted from events.

8. A — Structured fields make analytics and correlation more precise.

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10. A — Correlation combines multiple signals to identify suspicious or meaningful patterns.

Intermediate Level (10 Questions)

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A. Collecting and sending log data into a system

B. Deleting logs

C. Compiling code

D. Creating DNS records

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A. A named piece of extracted data

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C. A dashboard only

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A. To enable efficient filtering, aggregation, and search

B. To increase file size only

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49. Prometheus & Grafana

Category: Observability **Exam:** 30 Questions

Beginner Level (10 Questions)

1. What type of system is Prometheus?

A. Time-series monitoring and alerting system

B. Relational database only

C. Message broker

D. CI server

2. What language is used for Prometheus queries?

A. PromQL

B. SQLX

C. MetricQL only

D. GrafQL

3. What does Grafana primarily provide?

A. Visualization and dashboards

B. Container runtime

C. Database replication

D. Source control

4. What is a metric label?

A. A key-value dimension attached to a time series

B. A password

C. A dashboard title only

D. A log file

5. Why avoid unbounded labels?

A. They can cause high-cardinality memory and storage costs

B. They disable HTTPS

C. They prevent queries

D. They remove timestamps

6. What type of system is Prometheus is the best choice?

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Beginner Level Answer Key & Explanations

1. A — Prometheus stores and queries labeled time-series metrics.

2. A — PromQL is Prometheus's query language.

3. A — Grafana visualizes data from Prometheus and many other data sources.

4. A — Labels identify dimensions and enable filtering and aggregation.

5. A — High-cardinality labels create many unique series and can significantly increase resource usage.

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50. SRE & Observability

Category: Observability **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In SRE & Observability, which statement best describes a core concept?

A. The primary mechanism used by SRE & Observability

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with SRE & Observability?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a SRE & Observability component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

4. Which design goal is most important for a production SRE & Observability solution?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

5. What is a common reason to automate SRE & Observability operations?

A. Consistency and repeatability

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Beginner Level Answer Key & Explanations

1. A — The first option represents a central concept used in SRE & Observability. The other options are distractors unrelated to the core mechanism.

2. A — Versioning, testing, least privilege, and observability are broadly applicable engineering practices.

3. A — Start with evidence and recent changes before making disruptive modifications.

4. A — Production systems should balance reliability, security, maintainability, and observability.

5. A — Automation reduces manual variation and makes repeatable operations easier to test and audit.

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51. Splunk

Category: Observability **Exam:** 30 Questions

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52. Active Directory & LDAP

Category: Security **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In Active Directory & LDAP, which statement best describes a core concept?

A. The primary mechanism used by Active Directory & LDAP

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with Active Directory & LDAP?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a Active Directory & LDAP component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

4. Which design goal is most important for a production Active Directory & LDAP solution?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

5. What is a common reason to automate Active Directory & LDAP operations?

A. Consistency and repeatability

B. To eliminate all testing

C. To hide failures

D. To prevent versioning

6. In Active Directory & LDAP, which statement best describes a core concept is the best choice?

A. The primary mechanism used by Active Directory & LDAP

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

7. Which practice is generally recommended when working with Active Directory & LDAP?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

8. A production issue occurs in a Active Directory & LDAP component. What should you check first? Choose the most appropriate option.

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

9. Which design goal is most important for a production Active Directory & LDAP solution in a troubleshooting context?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

10. What is a common reason to automate Active Directory & LDAP operations?

A. Consistency and repeatability

B. To eliminate all testing

C. To hide failures

D. To prevent versioning

Beginner Level Answer Key & Explanations

1. A — The first option represents a central concept used in Active Directory & LDAP. The other options are distractors unrelated to the core mechanism.

2. A — Versioning, testing, least privilege, and observability are broadly applicable engineering practices.

3. A — Start with evidence and recent changes before making disruptive modifications.

4. A — Production systems should balance reliability, security, maintainability, and observability.

5. A — Automation reduces manual variation and makes repeatable operations easier to test and audit.

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53. Cybersecurity

Category: Security **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In Cybersecurity, which statement best describes a core concept?

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C. A hardware-only feature

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2. Which practice is generally recommended when working with Cybersecurity?

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D. Randomly change configuration

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C. No documentation

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A. Consistency and repeatability

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54. HashiCorp Vault

Category: Security **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In HashiCorp Vault, which statement best describes a core concept?

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2. Which practice is generally recommended when working with HashiCorp Vault?

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A. Logs, metrics, recent changes, and reproducible symptoms

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4. Which design goal is most important for a production HashiCorp Vault solution?

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55. Playwright & Cypress

Category: Testing **Exam:** 30 Questions

Beginner Level (10 Questions)

1. In Playwright & Cypress, which statement best describes a core concept?

A. The primary mechanism used by Playwright & Cypress

B. A filesystem permission unrelated to the technology

C. A hardware-only feature

D. A deprecated convention

2. Which practice is generally recommended when working with Playwright & Cypress?

A. Use version control, testing, and least privilege

B. Disable all logging

C. Hard-code every environment value

D. Avoid monitoring

3. A production issue occurs in a Playwright & Cypress component. What should you check first?

A. Logs, metrics, recent changes, and reproducible symptoms

B. Delete the component immediately

C. Disable security controls

D. Randomly change configuration

4. Which design goal is most important for a production Playwright & Cypress solution?

A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

5. What is a common reason to automate Playwright & Cypress operations?

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B. To eliminate all testing

C. To hide failures

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B. Maximum complexity

C. No documentation

D. Single points of failure

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A. Reliability, security, maintainability, and observability

B. Maximum complexity

C. No documentation

D. Single points of failure

5. What is a common reason to automate Playwright & Cypress operations?

A. Consistency and repeatability

B. To eliminate all testing

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6. In Playwright & Cypress, which statement best describes a core concept is the best choice?

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56. Selenium & Test Automation

Category: Testing **Exam:** 30 Questions

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